

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Third Semester B.E. (AIML/CCE) (Credit System) Degree Examinations**

April 2023

21AM303 / 21CC303 – DIGITAL SYSTEMS AND COMPUTER ORGANIZATION

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Design a logic circuit whose output is high only when majority of inputs A, B and C are low. And also implement it using only NAND gates.	8	L*3	1	2
	b) Realize all the Basic gates using NAND and NOR gate.	7	L2	1	1
	c) Mention and prove De-Morgan's Theorem.	5	L1	1	1
2.	a) Differentiate among multiplexer and demultiplexer using circuit diagram.	8	L2	2	1
	b) Using Kmap technique realize the reduced SoP and PoS form of the following function $f(a,b,c,d) = \sum m(1,2,3,5,6,7,13) + \sum d(8)$.	7	L3	1	2
	c) Solve $F(A,B,C,D) = \sum m(0,1,2,6,8)$ using Boolean law's.	5	L3	1	2
3.	a) Construct the following functions using one 4:1 multiplexer. $F(A,B,C,D) = \sum m(0,2,4,6,8,10,12,14)$.	7	L3	2	2
	b) Explain parity generator and checker.	7	L2	2	1
	c) Implement 2- Bit magnitude comparator.	6	L2	2	1
Unit – II					
4.	a) Describe in detail the interconnection of different components within the basic pressing circuit using single bus organization.	8	L2	3	1
	b) Explain addressing modes (any 6) with example.	7	L2	3	1
	c) Explain instruction execution and straight-line sequencing with example.	5	L1	3	1
5.	a) With neat diagram give the control sequence for the execution of the instruction MUL R ₁ , R ₂ , R ₃ using three Bus organization.	8	L2	3	1
	b) Explain with diagram organization of Microprogrammed control unit to support conditional branching.	7	L2	3	1
	c) Show the control sequence execution of a complete instruction Add (R ₃), R ₁ .	5	L2	3	1
6.	a) Using Booth's algorithm of multiplier, multiply +13 (multiplicand) with - 6 (multiplier).	10	L2	4	2
	b) Using Non Restoring Integer Division method to divide 11 from 5.	10	L2	4	2
Unit – III					
7.	a) Explain classic five stage pipeline for a RISC processor.	7	L2	5	1
	b) With neat sketch explain operand forwarding method to handle data hazards in H/W.	8	L2	5	1
	c) Discuss major hurdle of pipelining.	5	L2	5	1
8.	a) Discuss control dependency and its solution.	7	L2	5	1
	b) Derive an equation for speedup with pipelining.	5	L2	5	1
	c) Explain how to deal with branches in a pipeline.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
